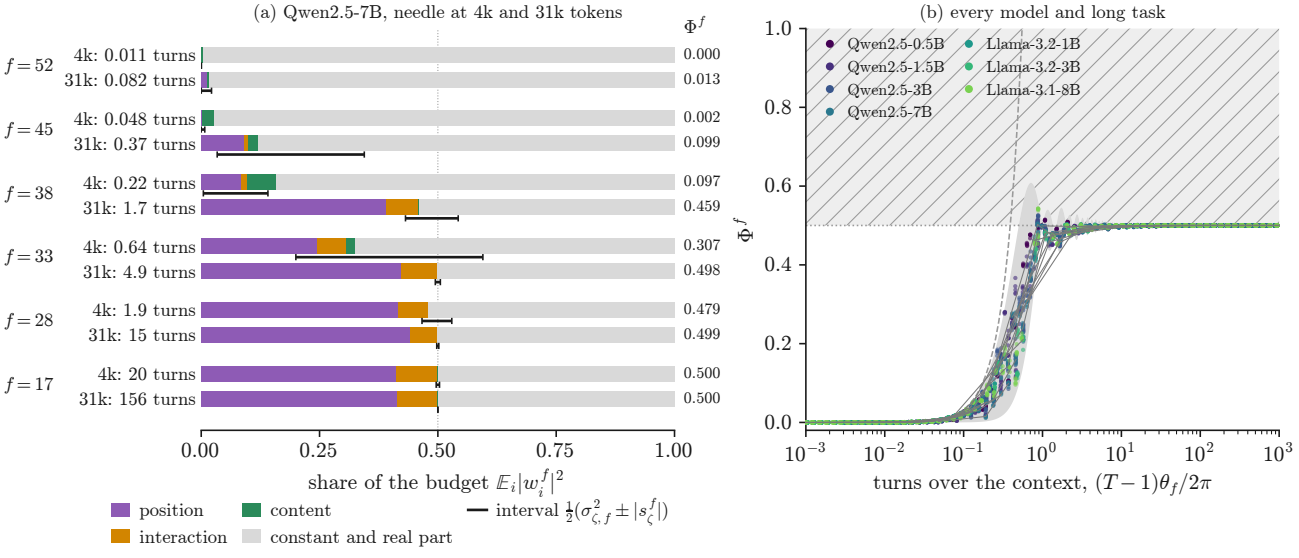

(a) Qwen2.5-7B, needle at 4k and 31k tokens
Φ^f
$f = 52$
4k: 0.011 turns
0.000
31k: 0.082 turns
0.013
$f = 45$
4k: 0.048 turns
0.002
31k: 0.37 turns
0.099
$f = 38$
4k: 0.22 turns
0.097
31k: 1.7 turns
0.459
$f = 33$
4k: 0.64 turns
0.307
31k: 4.9 turns
0.498
$f = 28$
4k: 1.9 turns
0.479
31k: 15 turns
0.499
$f = 17$
4k: 20 turns
0.500
31k: 156 turns
0.500
0.00
0.25
0.50
0.75
1.00
share of the budget $\mathbb{E}_i|w_i^f|^2$
position
content
interval $\frac{1}{2}(\sigma_{\zeta,f}^2 \pm |s_\zeta^f|)$
interaction
constant and real part
(b) every model and long task
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-1B
Llama-3.2-3B
Llama-3.1-8B
Φ^f
1.0
0.8
0.6
0.4
0.2
0.0
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
turns over the context, $(T-1)\theta_f/2\pi$